

AI-Driven Conversational Assistant for Leasing Company Customer Care

Group Id - 25-26J-179

Project Proposal Report

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B.Sc. (Hons) Degree in Information Technology
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Declaration

Machine Learning Application in Financial Leasing

Declaration

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The above candidates are carrying out research for the undergraduate Dissertation under my supervision.



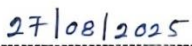
Signature of the Supervisor:
(Mrs. Suriya Kumari)



Date



Signature of the Co - Supervisor:
(Mrs. Dulani Jayasinghe)



Date

Abstract

Providing timely, accurate, and accessible customer support remains a significant challenge for vehicle leasing companies, particularly when handling a high volume of inquiries regarding leasing terms, company information, branch locations, and financial product details. Conventional customer service approaches that depend on human agents alone are often costly, limited in availability, and inconsistent in the quality of information delivered. This research proposes the development of an AI-driven conversational assistant specifically designed to support customer care operations within a leasing company environment.

The proposed system integrates a structured knowledge base stored and managed through Supabase with state-of-the-art Large Language Models (LLMs) accessed via the OpenRouter API to deliver accurate, context-aware responses across multiple interaction modalities. The assistant supports text-based, voice-based, and image-based queries, utilizing DeepSeek V3 for text conversations, Gemini 2.0 Flash for image understanding, and Deepgram for speech-to-text (STT) and text-to-speech (TTS) capabilities. Vector embedding techniques are employed to store knowledge base data, enabling semantic similarity search so that user queries are matched against the most relevant knowledge entries before generating a response.

The system is structured with two access levels: an administrative interface for managing and updating the knowledge base, and a user-facing chatbot interface for customers. Additionally, the assistant includes a live human agent handoff feature, allowing users to escalate complex or sensitive queries directly to a human agent, who can resolve the issue and close the chat session upon completion. The knowledge base architecture is specifically chosen to allow easy insertion, updating, and deletion of leasing-specific information, ensuring that the assistant remains current with evolving company data.

The expected outcome is a robust, multi-modal AI assistant capable of reducing response time, minimizing reliance on human agents for routine inquiries, and enhancing overall customer satisfaction for leasing company clients.

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List of Abbreviations

Table 1- Abbreviation

Abbreviation	Description
AI	Artificial Intelligence
LLM	Large Language Model
NLP	Natural Language Processing
RAG	Retrieval-Augmented Generation
STT	Speech-to-Text
TTS	Text-to-Speech
API	Application Programming Interface
UI	User Interface
KB	Knowledge Base
XAI	Explainable Artificial Intelligence
DB	Database
FAQ	Frequently Asked Question

1. Introduction

The vehicle leasing industry in Sri Lanka has experienced considerable growth in recent years, with an increasing number of individuals and businesses relying on leasing services for transportation and commercial mobility. As customer volumes expand, leasing companies face mounting pressure to deliver prompt, accurate, and consistent customer service across multiple channels. Currently, many leasing companies rely predominantly on human agents to handle customer inquiries, which presents challenges related to availability, scalability, and consistency of service quality.

Customers regularly seek information on topics such as leasing terms and conditions, monthly installment calculations, company branch locations, available vehicle models, application procedures, and general company background. When this volume of inquiries is managed solely through human agents, it leads to longer waiting times, higher operational costs, and a risk of inconsistent or outdated information being communicated to customers.

Recent advancements in Artificial Intelligence (AI), particularly in the domain of Large Language Models (LLMs) and Natural Language Processing (NLP), have opened new possibilities for automating customer interactions through intelligent conversational agents. AI-driven chatbots and virtual assistants have been widely adopted in the banking and financial services sectors globally, with demonstrated improvements in customer satisfaction and operational efficiency. However, the specific application of multi-modal AI assistants capable of handling text, voice, and image-based queries within Sri Lanka's vehicle leasing sector remains largely unexplored.

This component proposes to develop an AI-driven conversational assistant tailored to the needs of a leasing company's customer care operations. The system will integrate a curated knowledge base hosted on Supabase with multiple LLMs accessed through the OpenRouter API, including DeepSeek V3 for text-based conversations, Gemini 2.0 Flash for image-based queries, and Deepgram for speech recognition and synthesis. Vector embedding techniques will be employed to facilitate semantic similarity search, ensuring responses are contextually relevant and accurate.

The system will serve two user groups: customers who interact through a public-facing chatbot interface, and administrators who manage and update the knowledge base content. A human agent handoff feature will also be integrated to handle complex queries that require direct human assistance. This design ensures that the assistant supplements, rather than replaces, human expertise in customer care operations.

1.1 Background & Literature Survey

Customer service automation in financial services has been an active area of research and commercial development for over two decades. Early implementations relied on rule-based chatbot systems that could only respond to a predefined set of keywords or scripted conversation flows [1]. While these systems reduced basic workload for customer service teams, they were brittle and unable to handle varied natural language expressions of similar questions [5].

The emergence of transformer-based language models, beginning with BERT in 2018 and followed by the GPT series, has fundamentally transformed the capability of conversational AI systems. These models are trained on vast corpora of text and can generate contextually coherent responses to a wide range of natural language inputs without explicit programming for each scenario [3]. In the banking sector, institutions have adopted LLM-powered virtual assistants to handle account inquiries, loan status checks, and financial product explanations, achieving significant reductions in call center volumes [6].

Retrieval-Augmented Generation (RAG) represents a further refinement of LLM-based systems, combining the generative capability of language models with information retrieval from structured knowledge bases [7]. In this approach, user queries are converted into vector embeddings and compared against a database of pre-embedded knowledge entries using semantic similarity metrics. The most relevant knowledge entries are then provided as context to the LLM, which generates a response grounded in verified information rather than relying entirely on the model's training data. This technique is particularly important in regulated or domain-specific contexts, such as financial leasing, where accuracy and factual correctness are critical [8].

Multi-modal AI systems capable of processing text, voice, and image inputs have seen growing adoption across customer service platforms. Speech-to-text (STT) and text-to-speech (TTS) technologies have matured considerably, with systems such as Deepgram offering high-accuracy transcription and natural-sounding voice synthesis at scale [9]. Vision-language models such as Gemini 2.0 Flash enable chatbots to interpret image inputs, such as photographs of documents or leasing contracts, further broadening the scope of customer queries that can be handled automatically [10].

Despite this global progress, multi-modal AI customer care systems combining knowledge base retrieval with LLM generation have not been widely deployed in Sri Lanka's vehicle leasing sector. Most local leasing companies continue to depend on static FAQ pages and human agent support, leaving considerable room for improvement in service speed and consistency [4], [2].

1.2 Research Gap

A review of existing customer service AI implementations in the financial and leasing domain reveals several notable gaps that the proposed system seeks to address.

First, most deployed chatbot systems in Sri Lankan financial institutions rely on rule-based or intent-classification approaches that lack the flexibility to handle varied or complex natural language queries. These systems cannot generalize beyond the pre-programmed intent categories and frequently fail when customers phrase questions in unexpected ways.

Second, existing systems are predominantly text-only and do not accommodate voice-based or image-based interactions. For customers with limited literacy, accessibility challenges, or who simply prefer spoken interaction, text-only systems present a barrier to effective service.

Third, while knowledge base management systems exist in various commercial platforms, there is limited adoption of vector-embedding-based retrieval in local leasing company systems. Vector embedding enables semantic search that goes beyond keyword matching, allowing the system to retrieve relevant knowledge entries even when the user's phrasing does not exactly match stored content.

Fourth, no existing locally deployed system combines automated AI response with a seamless live human agent handoff, which is essential for handling queries that involve sensitive information, complaints, or issues requiring discretionary judgment.

Table 2 below summarizes these gaps in comparison to the proposed solution.

Table 2 - Research Gap

Feature / Criteria	Ref [12]	Ref [13]	Ref [15]	Ref [14]	Proposed Solution
LLM-powered conversational AI with knowledge base retrieval (RAG)	X	✓	X	X	✓
Multi-modal input support (text, voice, image)	X	X	X	X	✓
Vector embedding-based semantic search	X	X	X	X	✓
Real-time live human agent handoff	X	X	X	X	✓
Admin knowledge base management with easy update/delete	X	X	✓	X	✓

1.3 Research Problem

Vehicle leasing companies in Sri Lanka face increasing operational challenges in delivering consistent, accurate, and timely customer support. As customer inquiry volumes grow covering topics such as leasing eligibility, installment details, branch information, and company policies the reliance on human agents alone becomes unsustainable from both a cost and scalability perspective.

Existing automated systems in the domain are typically limited to rigid, rule-based chatbots that cannot adapt to varied natural language inputs, do not support voice or image-based interaction modes, and lack the ability to retrieve accurate, up-to-date company-specific information dynamically from a managed knowledge base. The absence of a semantic retrieval layer means that responses are either overly generic or potentially outdated, eroding customer trust and satisfaction.

Furthermore, no existing system in this context provides a seamless transition from AI-handled responses to live human agent support, leaving customers without recourse when the automated assistant cannot adequately resolve their query. This gap in hybrid AI-human service delivery represents a significant barrier to adoption of AI-based customer care in the leasing industry.

To address these challenges, this research proposes to develop an AI-driven conversational assistant that combines a structured, vector-embedded knowledge base with multiple LLMs to handle text, voice, and image queries from leasing customers. By integrating DeepSeek V3, Gemini 2.0 Flash, and Deepgram within a unified platform with an admin layer for knowledge base governance and a human agent handoff channel the proposed system aims to deliver accurate, accessible, and scalable customer support for vehicle leasing operations.

2. Objectives

2.1 Main Objective

The primary objective of this research is to design, develop, and deploy an AI-driven conversational assistant for a leasing company's customer care operations. The system will combine a dynamically managed knowledge base with multiple state-of-the-art Large Language Models to support multi-modal customer interactions including text, voice, and image-based queries and provide accurate, context-aware responses grounded in verified company information. The system will also incorporate a live human agent handoff capability for queries that require direct human intervention, and an administrative interface for continuous knowledge base maintenance.

2.2 Specific Objectives

- To collect, structure, and store leasing company knowledge base data including company background, branch details, leasing terms, and FAQs in a Supabase vector database using embedding techniques.
- To develop a text-based conversational AI module using DeepSeek V3 via the OpenRouter API, incorporating semantic similarity search against the knowledge base for accurate response generation.
- To develop an image-based query handling module using Gemini 2.0 Flash to interpret and respond to image inputs from customers.
- To implement voice-based interaction capabilities using Deepgram STT and TTS services, enabling customers to interact with the assistant through spoken language.
- To design and implement an administrative interface for managing knowledge base content, including adding, updating, and deleting records, with appropriate authentication and access controls.
- To integrate a live human agent handoff feature, allowing users to escalate queries to a human agent and enabling agents to close resolved chat sessions.
- To evaluate the system's accuracy, response relevance, and usability through testing against real leasing company customer inquiry scenarios.

3. Methodology

3.1 System Diagram

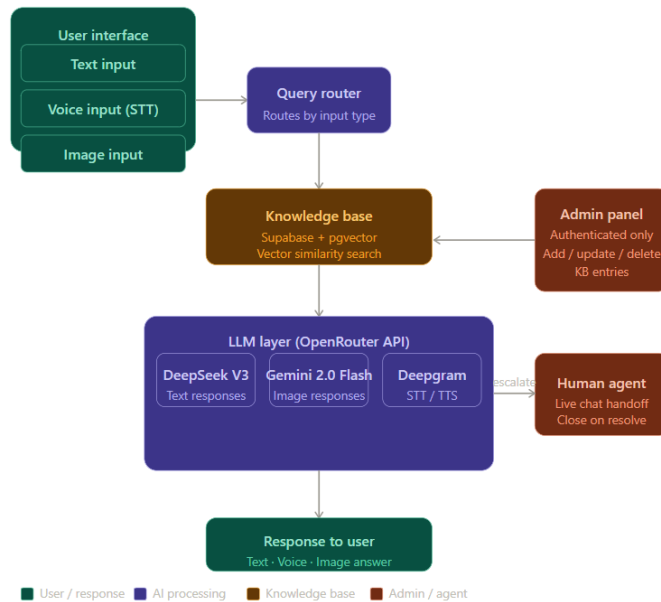


Figure 1- System Diagram

The system diagram presents a high-level architectural overview of the AI-Driven Conversational Assistant for Leasing Company Customer Care. The system is designed to handle multi-modal customer queries through an integrated pipeline that combines knowledge base retrieval with LLM-based response generation, and routes escalated queries to live human agents.

- **User Interface (Customer Chatbot):** The customer-facing interface supports three input modalities text, voice, and image. Customers can type queries, speak them aloud, or upload images relevant to their inquiry. The interface is accessible through the leasing company's website.
- **Query Router:** Incoming customer inputs are categorized by type (text, voice, or image) and routed to the appropriate processing module DeepSeek V3 for text, Gemini 2.0 Flash for images, and Deepgram STT for voice (converted to text before processing).
- **Knowledge Base (Supabase Vector Database):** Company-specific knowledge including branch details, leasing products, eligibility criteria, and company background is stored as vector embeddings in Supabase. When a query is received, a semantic similarity search retrieves the most relevant knowledge entries to serve as context for the LLM.
- **LLM Response Engine:** The retrieved knowledge context and the customer query are passed to the appropriate LLM, which generates a contextually accurate and natural-

language response. For voice queries, the text response is converted back to speech via Deepgram TTS.

- **Human Agent Handoff Module:** If a customer requests a live agent or if the AI cannot confidently resolve a query, the system routes the conversation to an available human agent. The agent can view the chat history, respond directly, and close the session upon resolution.
- **Admin Interface:** Authenticated administrators can add, update, or delete knowledge base entries through a dedicated management dashboard, ensuring the knowledge base remains current and accurate.

3.2 Gantt Chart

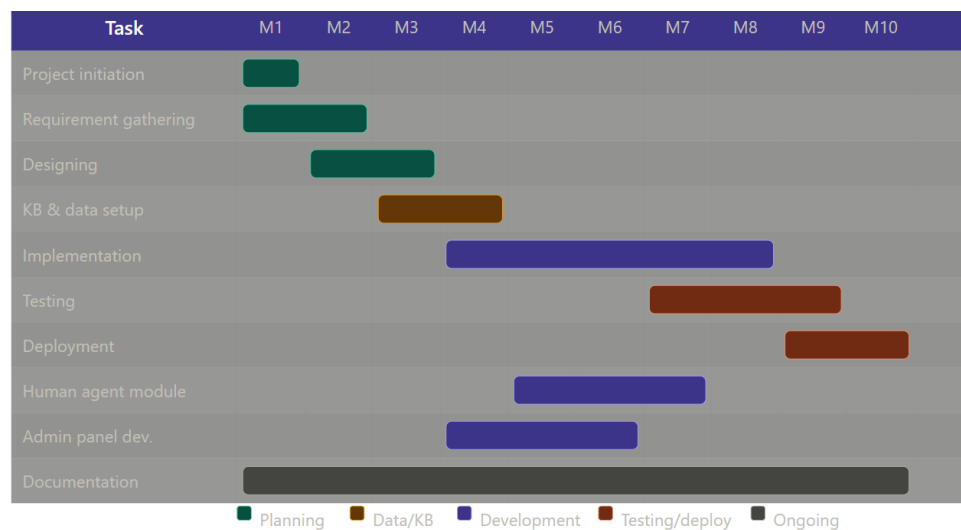


Figure 2- Gantt Chart

Table 3 - Gantt Chart

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3.3 Work Breakdown Structure

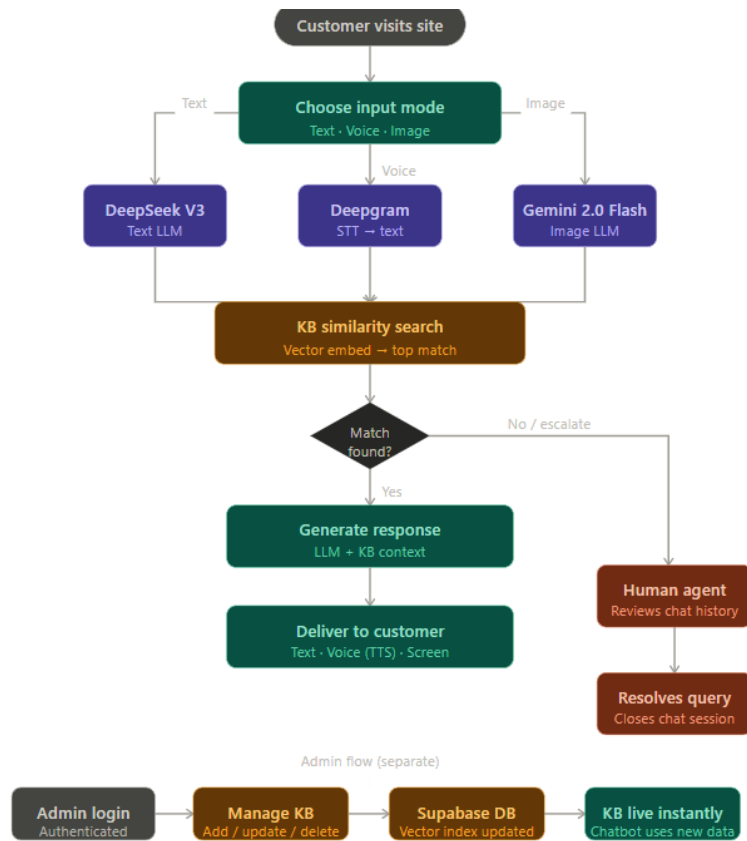


Figure 3- Work breakdown

4. Project Requirements

4.1 Functional Requirements

- Multi-Modal Customer Query Handling

The system must support text-based, voice-based, and image-based customer queries through a unified chatbot interface accessible on the leasing company's website. Text queries are processed by DeepSeek V3, image queries by Gemini 2.0 Flash, and voice queries via Deepgram STT for transcription and Deepgram TTS for spoken responses.

- Knowledge Base Retrieval

Upon receiving a customer query, the system must convert the query into a vector embedding and perform a semantic similarity search against the Supabase vector database. The most contextually relevant knowledge entries must be retrieved and supplied as context to the LLM for grounded response generation.

- LLM-Powered Response Generation

The system must leverage the OpenRouter API to access DeepSeek V3 for text conversations and Gemini 2.0 Flash for image-based queries. Responses must be accurate, contextually relevant, and expressed in natural language appropriate for customer-facing communication.

- Live Human Agent Handoff

The system must include a feature enabling customers to request a live human agent at any point during a chat session. The system must route the conversation to an available agent, provide the agent with the full chat history, and allow the agent to close the session once the query is resolved.

- Admin Knowledge Base Management

Authenticated administrators must be able to add new knowledge entries, update existing entries, and delete outdated records through a dedicated admin interface. All changes must be reflected immediately in the vector database to ensure up-to-date responses.

- Authentication & Access Control

The admin interface must be accessible only to authenticated users. Customer-facing chatbot access does not require authentication. Role-based access ensures that knowledge base management functions are restricted to authorized personnel only.

4.2 Non-Functional Requirements

- Performance

The system must generate responses to customer queries within 3, 5 seconds per request under normal operating conditions. Voice query transcription and synthesis must operate with minimal latency to ensure a natural conversational experience.

- Scalability

The system must handle concurrent customer sessions across the leasing company's website without degradation in response quality or speed. The knowledge base architecture must support growth in the volume of stored entries over time.

- Security

All data transmissions must be encrypted using TLS. Admin authentication must use secure credential management. The Supabase database must enforce row-level security policies to prevent unauthorized access to knowledge base data.

- Reliability & Availability

The customer-facing chatbot must maintain a minimum uptime of 99%. Fallback mechanisms must be in place to redirect users to a human agent or an error message in the event of LLM API unavailability.

- Maintainability

The system must be built using modular, well-documented code to enable future developers to update LLM integrations, add new query modalities, or expand the knowledge base schema with minimal effort.

- Usability

The customer chatbot interface must be intuitive and accessible for users with varying levels of digital literacy. The admin panel must provide clear navigation for knowledge base management operations without requiring technical expertise.

4.3 User Requirements

- Accessible and Intuitive Interface

Customers must be able to interact with the AI assistant through a clearly designed chatbot interface on the leasing company website, with visible options to switch between text, voice, and image input modes.

- Accurate and Relevant Responses

The system must provide reliable responses drawn from verified company knowledge base data, covering leasing terms, branch information, company background, and frequently asked questions.

- Voice Interaction Support

Users must be able to speak their queries and receive audible responses, enabling interaction for users who prefer or require voice-based communication.

- Image Query Support

Users must be able to upload images such as photographs of documents, vehicles, or leasing materials and receive relevant AI-generated responses.

- Human Agent Escalation

Users must have a clearly visible option at any point to request connection with a live human agent, with confirmation that the escalation has been initiated.

4.4 System Requirements

Table 4 - Tools and Technology

Tools & Technologies	Purpose
Frontend Development (nextjs)	Building interactive chatbot UI and admin dashboard
Next js	Structuring, styling, and scripting the frontend application
Backend Development (Next.js)	Developing APIs for query routing, LLM integration, and KB management
REST API	Communication between frontend, backend, and external AI services
Supabase (PostgreSQL + pgvector)	Vector database for knowledge base storage and semantic similarity search
OpenRouter API	Accessing multiple LLM models including DeepSeek V3 and Gemini 2.0 Flash
DeepSeek V3	Text-based conversational AI response generation
Gemini 2.0 Flash	Image-based query understanding and response generation
Deepgram	Speech-to-text (STT) and text-to-speech (TTS) voice capabilities
Vector Embedding (text-embedding models)	Encoding knowledge base entries and user queries for semantic search
Git & GitHub	Version control and collaborative development

Tools & Technologies	Purpose
Ms planner	Task tracking, sprint planning, and project management
Microsoft Teams	Real-time communication and project collaboration
Ms Word	Report writing, documentation preparation

5. Budget and Budget Justification

Table 5- Budget table

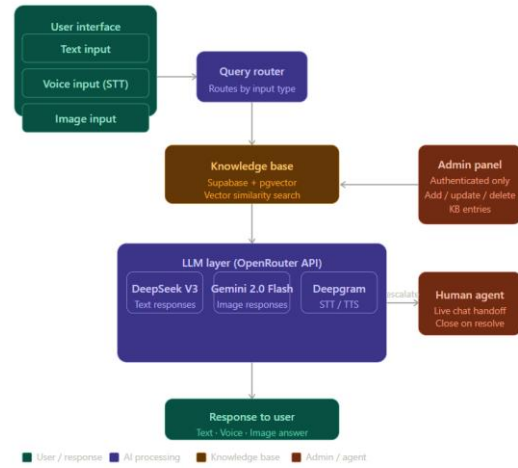
Item	Cost	Justification
Computing Resources (Local / University Lab)	Free	Use existing university lab PCs/servers; no additional cost required.
Software & Tools	Free	Open-source and free-tier tools: React, Node.js, Supabase free tier, OpenRouter API free model access, Deepgram free tier.
LLM API Usage (OpenRouter - Free Models)	Free	DeepSeek V3 and Gemini 2.0 Flash are available via OpenRouter's free-tier access for development and testing.
Knowledge Base Development	Free	Research assistant effort to structure, embed, and load leasing company knowledge entries into Supabase.
System Development & Integration	\$0.15	Effort for frontend, backend, and multi-modal AI integration development and testing.
Documentation & Reporting	Rs. 2,500	Printing, thesis preparation, and minor documentation costs.
Contingency (10%)	Rs. 650	For any minor unforeseen research-related costs.

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7. Appendices

- System Diagram



- Work Breakdown Structure

